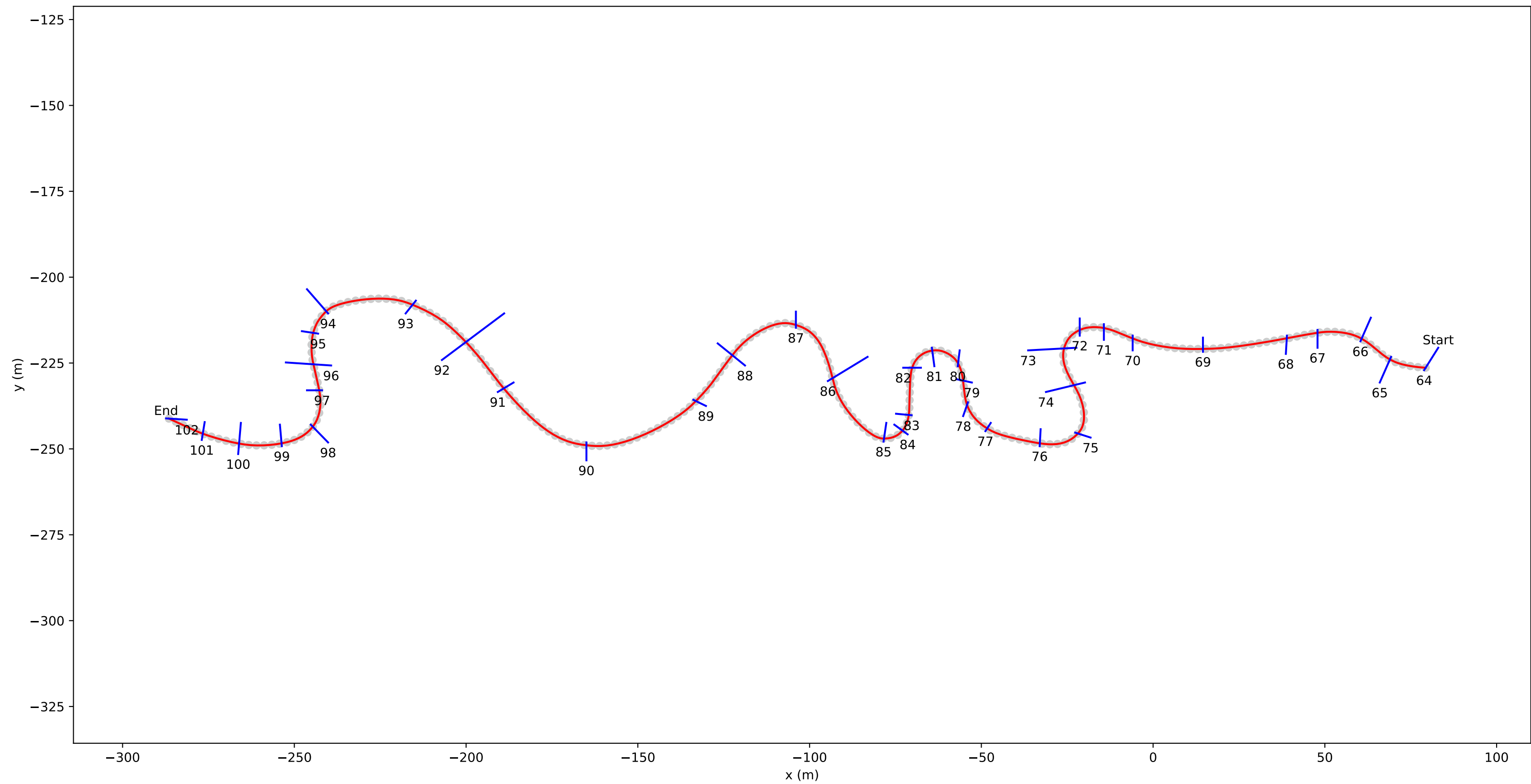


Map plot: optimized path through gates



Track Generation Report

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Generated: 2026-08-23 02:14:50

Gate file: ref_less_gates_twist.csv

Gate count: 39

Saved DOFs file: opt_dofs_2026-08-23_02-14-50.npy

Saved DOF history file: dof_history_2026-08-23_02-14-50.npy

Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

Point counts

path generation points: 234

Results

total optimization time: 3286.147 s

total objective function evaluations: 45176

optimizer iterations (accepted steps): 96

total path length: 521.4182 m

total objective function value: 1.870537

Per-step time estimate (one measured objective evaluation)

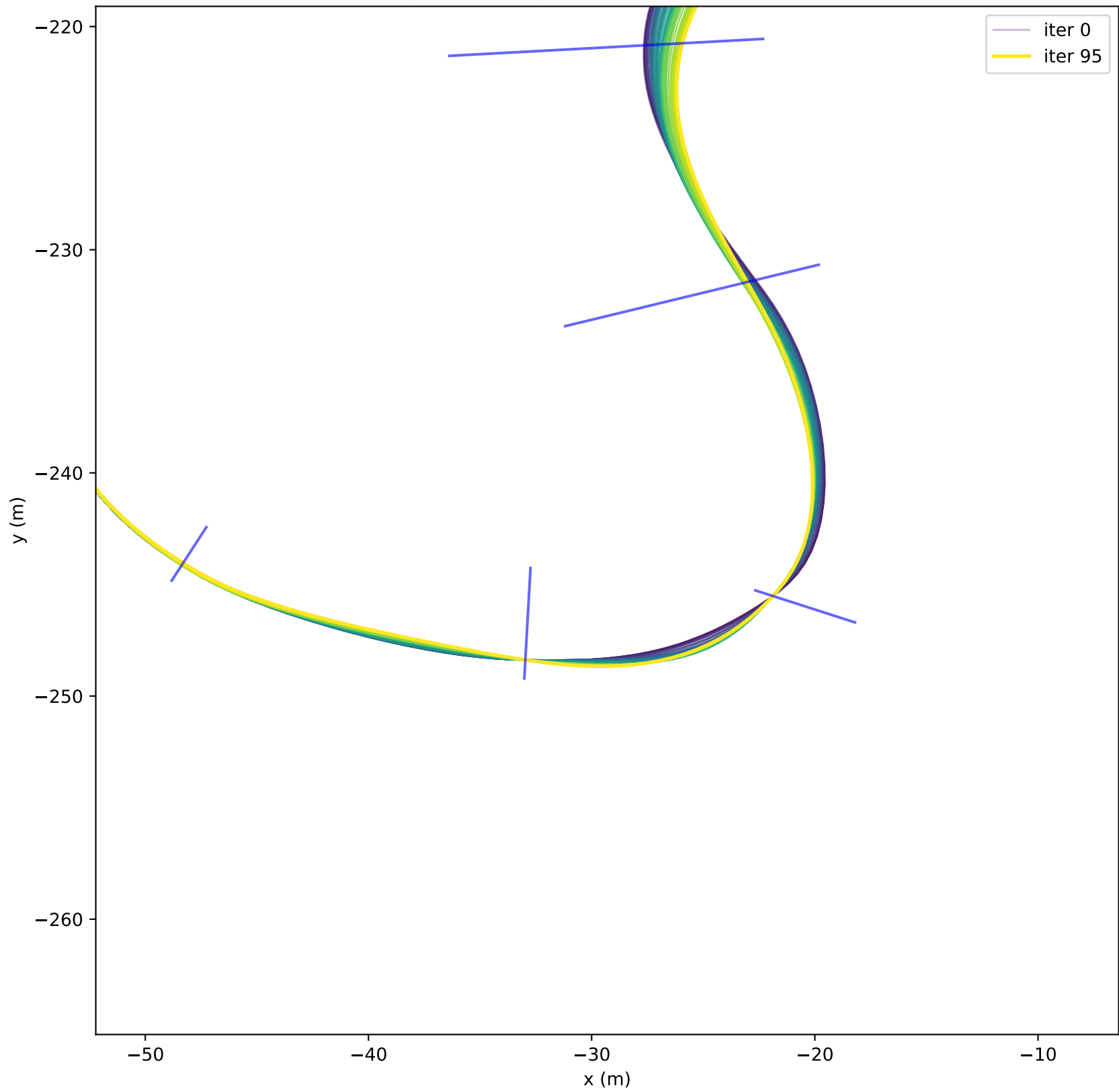
make spline from dofs: 5.42 ms

path length function: 0.42 ms

curvature objective function: 73.96 ms

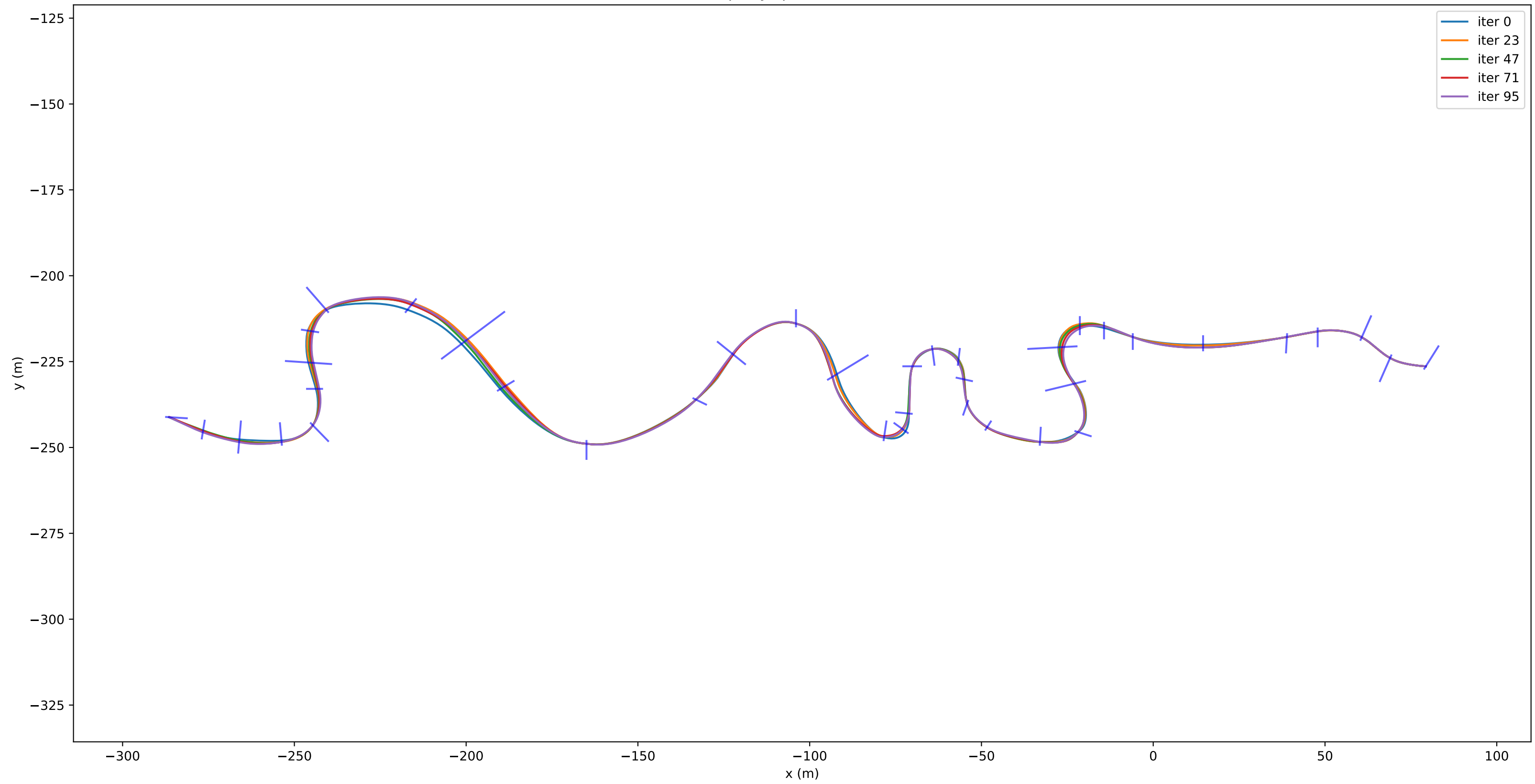
Optimization progression

Optimization progression (96 iterations)

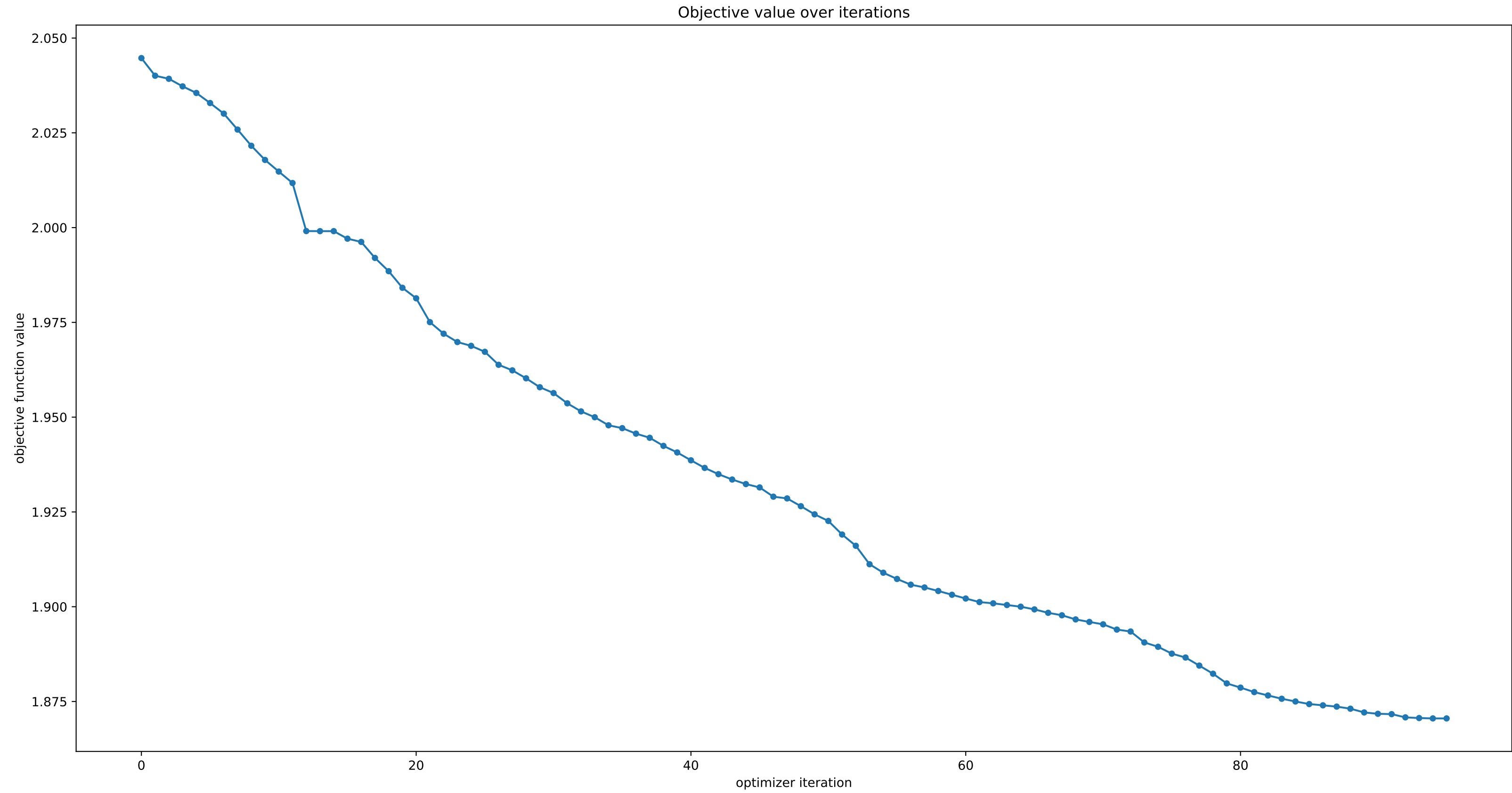


Equally spaced iterations

5 equally spaced iterations



Objective value over iterations



Instant radius vs distance

